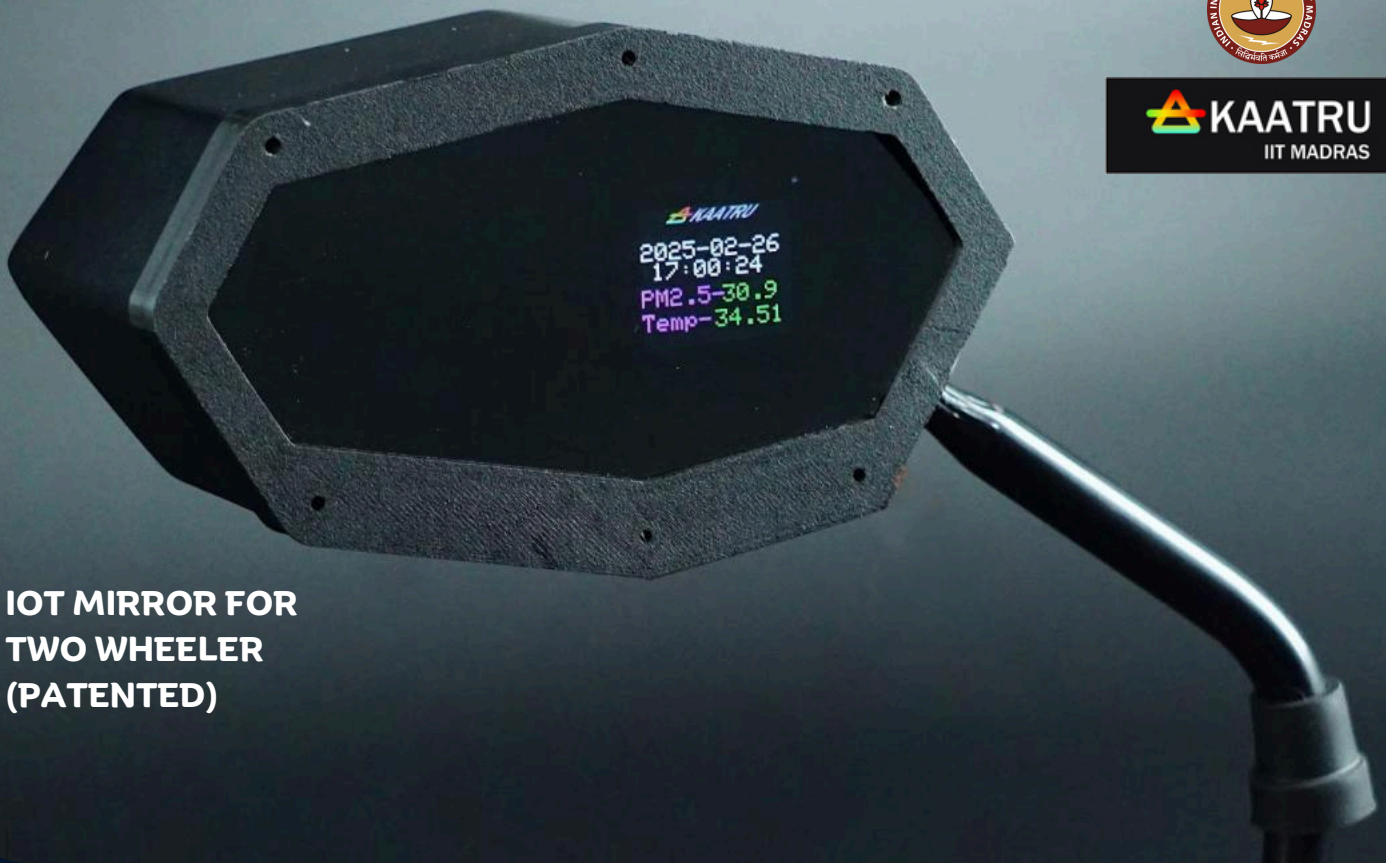




KAATRU
IIT MADRAS



**IOT MIRROR FOR
TWO WHEELER
(PATENTED)**

MOBILE MONITORING FOR HYPERLOCAL AIR QUALITY INSIGHTS

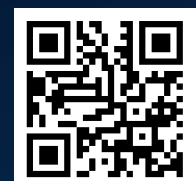


**MOBILE SENSOR
FOR FOUR
WHEELERS**



**STATIONARY MULTI
PARAMETER
MONITORING**

COLLABORATORS



RAGHUR@IITM.AC.IN

ABOUT US

Air pollution remains a profoundly significant issue in our nation, with India ranking third globally in environmental contamination in 2023, recording an average AQI of 147. Possessing the largest population among critically polluted nations nearly ten times that of Bangladesh, the most affected India was home to 65 of the 100 most pollution-intensive cities worldwide. Furthermore, in 2019, deteriorating air quality was a contributing factor to over 1.6 million fatalities across the country. It has been linked to various health problems such as respiratory diseases, allergies, and even cancer.

Hence, the systematic monitoring of air quality is imperative. Presently, India relies on a network of only 563 CAAQM stations, yet the data they provide is often inconsistent, resulting in significant discrepancies in AQI measurements. To enhance precision and reliability, a paradigm shift from stationary monitoring to dynamic hyperlocal mobile based air quality assessment is warranted. Moreover, the condition of roads in the city also contributes to ambient air pollution. Poorly maintained roads lead to more dust and vehicle emissions, increasing the area's pollution levels.

We are a research group named “KAATRU” guided by Dr Raghunathan Rengaswamy Marti Mannariah Gurunath Institute Chair Professor, Dean of Global Engagement Office, and the head of Walmart Centre for Tech Excellence (WCTE) at IIT Madras. The team is leveraging IoT and Data Science to deliver high spatio-temporal resolution insights on air quality and road conditions in cities. The project aims to support government bodies, regulators, manufacturers, and the general population with nuanced insights for strategic decision-making.

Pilot study initiatives were executed across three Indian cities, with Chennai serving as a focal point for extensive ambient air pollution assessments. In collaboration with Centre for Urbanization, Building & Environment (CUBE), a comprehensive monitoring study was conducted in the Ambattur area, encompassing residential, industrial, and school zones. Additionally, a Diwali-centric air quality analysis was carried out across Velachery, IIT Madras, Guindy, and Saidapet, capturing pre-Diwali, Diwali, and post-Diwali pollution levels (2019) and mapping the entire Koyambedu market area's pollution level (2024).

Further studies in Chennai included an assessment of road conditions and air quality citywide in partnership with NSE Foundation (2023), the development of an alternate route optimization map identifying the least polluted and least bumpy pathways between IIT Madras and Triplicane (2022), and an internal road condition mapping project within IIT Madras (2023). Moreover, an ongoing collaborative effort with Dept of civil engineering, IIT Madras and Princeton University (2025) is focused on real-time air quality data monitoring for the city.

In Gurugram, an extensive mapping initiative is being conducted in collaboration with NSE Foundation and THSTI to examine the impact of air quality on pregnancy outcomes and infant growth, with data collection efforts ongoing since 2023. Additionally, a partnership with Walmart Centre for Tech Excellence(2025) aims to analyze indoor air quality indices (AQI) within MSME foundry and manufacturing industries, providing critical insights into occupational environmental conditions. Our initiative delivers hyperlocal intelligence across multiple domains, encompassing Business-to-Government (B2G), Business-to-Business (B2B), and Business-to-Customer (B2C) interactions, thereby fostering informed decision-making and enhanced operational efficacy.

SERVICES

BUSINESS TO GOVERNING MODEL

We provide an advanced data-driven framework for pollution mitigation and infrastructure optimization. The Pollution Control Board can leverage real-time AQI index, geospatial hotspot detection, and emission source attribution to enable precise environmental interventions. Road authorities such as NHAI can implement AI-powered pavement diagnostics to facilitate predictive maintenance and enhance infrastructure longevity. Furthermore, dynamic speed modulation, calibrated to road topology and traffic patterns, will improve vehicular safety, minimize mechanical degradation, and mitigate congestion, fostering a more sustainable and resilient urban mobility ecosystem.

BUSINESS TO BUSINESS MODEL

Vehicle aggregators leveraging advanced fleet management solutions enhance premium ride experiences by optimizing ride quality and travel time through intelligent Estimated Time of Arrival predictions. It employs AI-driven driver-vehicle matching, factoring in driver profiles and vehicle conditions to align with customer preferences. GPS-backed data ensures seamless cab tracking, even during communication lapses, while predictive analytics optimize vehicle distribution based on road conditions. Weather anomalies, by integrating GPS-tagged emissions data over conventional pollution board inputs, refine surge pricing strategies and ensure regulatory compliance with precision.

Micro, Small, and Medium Enterprises (MSMEs) can leverage advanced indoor air quality mapping to ensure optimal environmental conditions, enhancing workforce well-being and productivity. By integrating intelligent ventilation optimization and automated maintenance scheduling, enterprises can preemptively mitigate inefficiencies and equipment degradation. Furthermore, automated compliance reporting streamlines regulatory adherence, minimizing administrative burden while ensuring operational excellence.

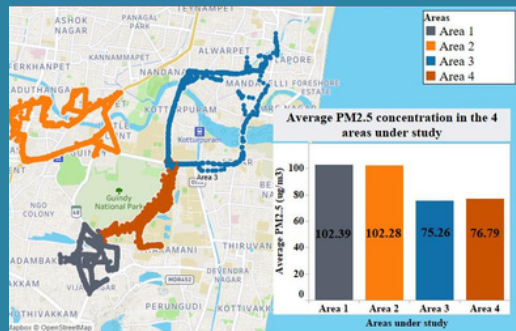
BUSINESS TO CUSTOMER MODEL

A Side view mirror (Two wheeler) and Mobile variant (four wheeler) optimizes vehicular navigation by identifying the least exposure route and tailoring the least bumpy path for enhanced ride comfort. It facilitates early vehicle failure detection, minimizes wear and tear, and informs drivers of impending road anomalies via terrain mapping for proactive maneuvering. Leveraging vehicle diagnostics and driver profiles, it enables predictive maintenance, ensuring optimal servicing and superior passenger ride quality.

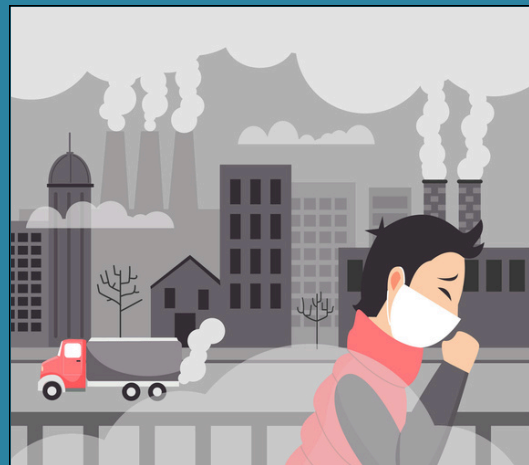
CONCLUSION

This initiative redefines air quality assessment in India by transitioning from traditional stationary monitoring to a contemporary mobile monitoring framework. Leveraging high spatio-temporal resolution analytics, it facilitates comprehensive environmental intelligence on urban air quality and road conditions. By equipping governmental agencies, regulatory bodies, industry stakeholders, and the general population with granular, data-driven insights, the project fosters informed policy formulation, strategic urban planning, and enhanced environmental governance.

SOLUTIONS



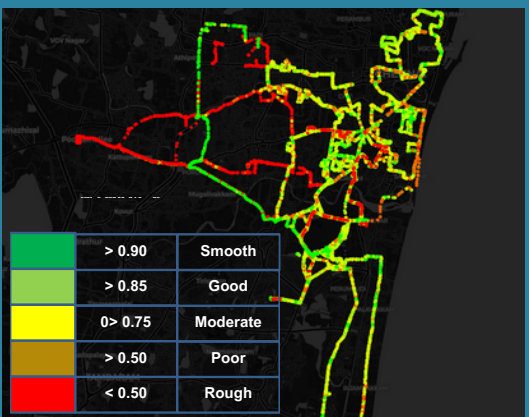
CLEANEST ROUTE



PERSONAL EXPOSURE ASSESSMENT

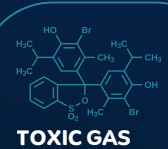


DATA DRIVEN POLICY AND MITIGATION STRATEGIES



ROAD CONDITION MONITORING

PARAMETERS



TOXIC GAS

0 - 1000 ppb
0 - 10000 ppm (CO₂)

PM2.5

PM

0 - 1000 $\mu\text{g}/\text{m}^3$
+/- 10 %
PM1.0, PM2.5, PM10



VOC

0 - 1000 ppm
< 10 s



AMBIENT LIGHT

UV
IIR
VISIBLE



IMU

ACCELEROMETER
GYROSCOPE
MAGNETOMETER
± 2g ± 4g ± 8g ± 16g



HUMIDITY

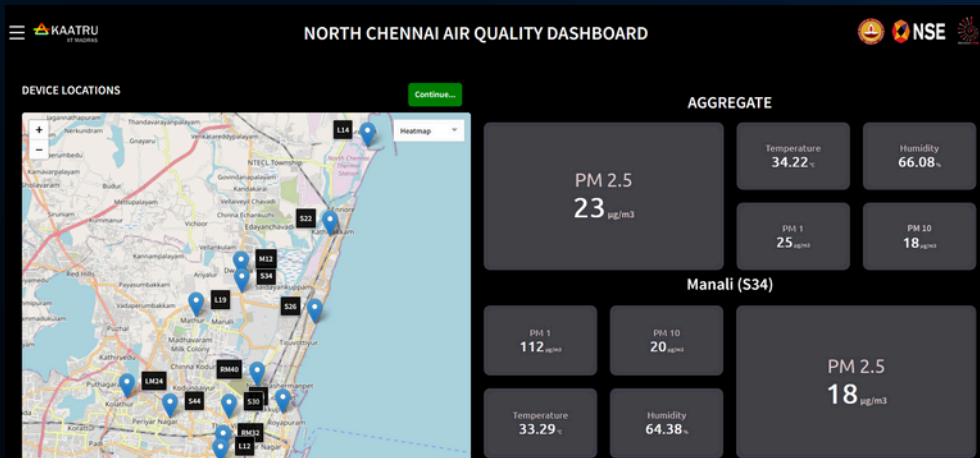
0 - 90 %RH
⌚ 20ms



TEMPERATURE

10 - 50 °C
⌚ 2

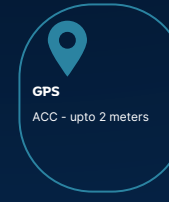
REALTIME AIR QUALITY DASHBOARD



SENSOR NETWORK DIAGNOSTICS



KEY FEATURES



**MOBILE APP
FOR AIR QUALITY UPDATES** ❄️ ❄️

